

$$a) \quad \frac{1}{5} + \frac{1}{5} \times \frac{1}{2}$$

$$= \frac{1}{5} + \frac{1}{10}$$

$$= \frac{2}{10} + \frac{1}{10}$$

$$= \frac{3}{10}$$

$$b) \quad \frac{11}{15} \div \frac{4}{5} - \frac{3}{4}$$

$$\frac{11}{15} \times \frac{5}{4} - \frac{3}{4}$$

$$\frac{11}{12} - \frac{3}{4}$$

$$\frac{11}{12} - \frac{9}{12}$$

$$\frac{2}{12} = \frac{1}{6}$$

$$c) \quad \left(\frac{3}{7} + \frac{1}{14} \right)^2$$

$$= \left(\frac{6}{14} + \frac{1}{14} \right) \times \left(\frac{6}{14} + \frac{1}{14} \right) = \frac{1}{4}$$

$$= \left(\frac{7}{14} \right) \times \left(\frac{7}{14} \right)$$

$$= \left(\frac{1}{2} \right) \times \left(\frac{1}{2} \right)$$

$$d) \frac{5}{12} - \frac{7}{13} \div \frac{2}{18}$$

$$= \frac{5}{12} - \frac{7}{13} \times \frac{26}{3}$$

$$= \frac{5}{12} - \frac{14}{3}$$

$$= \frac{5}{12} - \frac{56}{12}$$

$$= -\frac{51}{12}$$

$$= -\frac{17}{4}$$

$$e) \frac{11}{14} + \frac{3}{4} \times \frac{20}{21} + \frac{15}{8}$$

$$= \frac{11}{14} + \frac{5}{7} + \frac{15}{8}$$

$$= \frac{44}{56} + \frac{40}{56} + \frac{105}{56}$$

$$= \frac{189}{56} = \frac{27}{8}$$

$$f) \frac{12}{5} \div \frac{4}{5} - \frac{29}{20} \div \frac{3}{10}$$

$$= \frac{12}{5} \times \frac{5}{4} - \frac{29}{20} \times \frac{10}{3}$$

$$= \frac{27}{5} - \frac{29}{6}$$

$$= \frac{162}{30} - \frac{145}{30}$$

$$= \frac{17}{30}$$

$$9) 3\frac{1}{5} + 1\frac{7}{12} \times 3\frac{3}{5}$$

$$= \frac{16}{5} + \frac{19}{12} \times \frac{18}{5}$$

$$= \frac{16}{5} + \frac{57}{10}$$

$$= \frac{32}{10} + \frac{57}{10}$$

$$= \frac{89}{10}$$

$$4\frac{2}{5} \div \left(2\frac{1}{3} - \frac{33}{12} \right) \div 3\frac{1}{5}$$

$$= \frac{22}{5} \div \left(\frac{11}{5} - \frac{11}{4} \right) \div \frac{16}{5}$$

$$= \frac{22}{5} \div \left(\frac{44}{20} - \frac{55}{20} \right) \div \frac{16}{5}$$

$$= \frac{22}{5} \div \left(-\frac{11}{20} \right) \div \frac{16}{5}$$

$$= \frac{22}{5} \div \left(-\frac{20}{11} \right) \div \frac{16}{5}$$

$$= -8 \times \frac{5}{16}$$

$$= -\frac{5}{2}$$